

CLAY MINERALOGY OF BEDLOADS FROM MORANG RIVER, DRAINING RAJMAHAL BASALT: SIGNIFICANCE TO CATCHMENT WEATHERING PROCESS

Project submitted to the Central University of Punjab

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BY

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CERTIFICATE

I declare that the project entitled “**CLAY MINERALOGY OF BEDLOADS FROM MORANG RIVER , DRAINING RAJMAHAL BASALT : SIGNIFICANCE TO CATCHMENT WEATHERING PROCESS**” has been prepared by me under the guidance of Dr. Jitendra Kumar Pattanaik, Associate Professor, Department of Geology , School of Environment and Earth Sciences, Central University of Punjab. No part of this project has formed the basis for the award of any degree or fellowship previously.

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This is to certify that the work incorporated in this M.sc thesis entitled “**CLAY MINERALOGY OF BEDLOADS FROM MORANG RIVER , DRAINING RAJMAHAL BASALT : SIGNIFICANCE TO CATCHMENT WEATHERING PROCESS**” submitted by Mrs. **Sree Sruthy M S** to the Central University of Punjab in fulfillment of the requirements for the award of the Degree of **M.Sc.** Earth and Geological Science, under my guidance. I further certify that this work has not been submitted to any other University or Institution in part or full for the award of any degree or diploma. Research material obtained from other sources has been duly acknowledged in the thesis. Any text, illustration, table etc., used in the thesis from other sources, has been duly cited and acknowledged.

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ABSTRACT

**“CLAY MINERALOGY OF BEDLOADS FROM MORANG RIVER , DRAINING
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Mineralogical study of bed load sediments will help to understand the weathering process, the intensity of weathering and the denudation in various environmental conditions, origin, mode of occurrence, biological and physic-chemical processes, mobilization and transport of various elements. River draining a single lithology helps to study above parameter of the catchment area. So here Morang river draining Rajmahal basalt was selected as there is no report or literature available from this catchment to understand the weathering process in its water shade. The clay mineral compositions of the river bed load sediments from different locations shows quartz, calcic plagioclase, clinoenstatite clay minerals such as vermiculite, kaolinite, illite, goethite and clay micas, are dominating mineral phases. Clay mineral percentage is higher in the upper reaches compare to the lower reach of the river. Clay mineral gibbsite and kaoline indicate the removal of silica is higher in the upper reach and shows kaolinitization, lateritization and upto some degree of Bauxitization process is active in the upper reach. There occurs kaolinitization , lateritization and Bauxitization process.

(Name and signature of student)
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CHAPTER 1

INTRODUCTION

Clay minerals are hydrous aluminium phyllosilicates having particle size smaller than two microns. It exhibit colloidal behaviour, high dry strength, slow dilatancy and plasticity. Depending on the species, clay crystals are primarily composed of aluminium or magnesium, oxygen, silicon and hydroxyl (OH), along with a variety of related cations. Weathering of primary minerals produces clay minerals.

Weathering is the principal process responsible for the breakdown of rocks at the Earth's surface, by the activity of water, extremes of temperature, and natural action. It doesn't include the expulsion of rock material. There are three major types of weathering physical, chemical and biological weathering (Weaver (1980)).

(a) Physical weathering: physical disintegration of rock mass due to variation of temperature, mechanical properties of rock, change in over load, development of new minerals in fracture, mechanical abrasion due to different geological agents.

- i. Abrasion weathering: When rock or other features of Earth's surface are broken down into smaller pieces by forces like wind, water, and even glaciers, it is called abrasion weathering (Grim (1968)). These forces can cause abrasion by grating against rock fragments, known as clasts.
- ii. Exfoliation weathering occurs most often with igneous rocks. Just as exfoliating your skin involves rubbing off dead layers, exfoliation of rocks involves the flaking of outer layers of rocks. As these layers on the outside of a rock flake or fall off, a process known as unloading, pressure is released on the rock layers below.
- iii. Frost wedging occurs when liquid water in the pores of a rock freezes and expands, causing the rock to crack. When the ice melts again, the water can flow into new pores. As the freezing and melting process repeats, the cracks in rocks from frost wedging can become quite large.
- iv. Salt crystallization occurs in coastal and desert regions when water evaporates, leaving salt crystals behind. As these crystals expand, they exert

pressure on surrounding rock, eventually breaking the rock and leaving holes in the rock's surface. This process is also called haloclasty and the resulting rock surfaces have a honeycomb appearance. Spray from ocean water along the coast can cause this type of weathering (Weaver and Pollard (1973)).

- v. Thermal expansion, minerals expand and contract when heat fluctuates. When heat is lost, molecules contract and take up less space. As heat is added, molecules expand. As thermal expansion repeats, it causes stress on rocks and eventually they will crumble.
- vi. Root wedging, roots from large plants grow into small breaks in rocks. As the roots grow larger, they can cause the rocks to crack and break further.

(b) Chemical weathering: chemical weathering is the weakening and subsequent decomposing of primary mineral present in the rock by chemical processes. These processes include oxidation, hydrolysis, and carbonation.

Parent rock + ion-depleted water $\longrightarrow$ weathering complex + ion-enriched water

The weathering complex changes over time and produces soil as the natural transformation between the parent rock and the environment. The majority of weathering complexes and soils are mostly made up of clay minerals.

According on the make-up of the assault solution, chemical weathering comes in four various flavours.

(1) In organic and acid settings, where organo-mineral compounds are easily formed, acidolysis occurs.

(2) Salinolysis is a technique used to describe evaporitic saline environments (Na, K, etc.) in pH-normal settings.

(3) Alcalinolysis refers to fundamental circumstances and to water that has been charged with Ca or Mg.

(4) Hydrolysis is the process of attacking rocks with water that is only slightly ionised and at a medium pH. The most sophisticated and well-known method is hydrolysis. Extended data on weathering are given by Millot (1964, 1970), Loughnan (1969), Carroll (1970), Duchaufour et al. (1977) Drever (1985), Colman and Dethier (1986).

- i. Oxidation is the response of rock minerals with oxygen, in this way changing the mineral structure of the rock. At the point when minerals in rock oxidize, they turn out to be less resistant to weathering (Krishnaswami and Singh, 2005).
- ii. Carbonation is the procedure of minerals responding with carbonic acid. Carbonic acid is shaped when water consolidates with carbon dioxide. Carbonic acid disintegrates or separates minerals in the rock.



- iii. Hydrolysis is held to cause by water. Water changes the composition of mineral and size of minerals in rock making them less impervious to weathering. At the point when this mineral is totally hydrolyzed, earth minerals and quartz are created and such components as K, Ca or Na are discharged (Dixon and Weed (1977)).



- iv. Hydration is the assimilation of water into the mineral structure. A decent case of hydration is the retention of water by anhydrite, bringing about the development of gypsum. Hydration extends volume and furthermore brings about rock distortion.
- v. Dehydration is the expulsion of water from mineral structures. A decent case of lack of hydration is the expulsion of water from limonite, bringing about the development of hematite.

(c) Biological weathering: Biological weathering is the weakening and consequent breaking down of rock by plants, creatures and microorganisms. Developing plant roots can apply pressure or weight on rock. Natural procedures can likewise create chemical weathering, for instance where plant roots or microorganisms deliver natural acids which help to break up minerals. Microbial movement separates minerals by adjusting the substance piece, along these lines making it more to weathering. One case of microbial action is lichen, lichen is growths and green growth, living respectively in an advantageous relationship. Organisms discharge chemicals that separate minerals; the minerals hence discharged from rock are consumed by the green growth. As this procedure proceeds with, openings and

holes keep on developing on the rock, uncovering the rock further to physical and chemical weathering.

There are different factors which affect the weathering rate and formation of clays.

i) Climate: Chemical weathering affects warm regions, but physical weathering affects cold temperatures (particularly by frost action). Weathering is more pronounced in either situation as moisture content increases. Chemical reactions are more likely to occur quickly at higher temperatures. Most chemical reactions require water, which is a reactant in hydration and carbonation. Water is another medium in which the reaction might occur. Hot moist temperatures promote biological activity. In reality, a hot, humid climate is optimal for rapid chemical weathering.

Physical weathering by frost action is most likely in cold climates where freeze and thaw occur alternately during the cold weather. Again, precipitation is the key component in this scenario. Ice cannot form in the absence of water, and frost action is not conceivable. As a result, effective frost action occurs in the cold, damp climate.

ii) Particle Size: The size of rock particles influences the rate of chemical weathering. Weathering is accelerated when the rock bits are smaller. This is because the surface area exposed to weather activity is greater when the rock particles are smaller.

iii) Exposure: The level of exposure, or the extent to which the rock comes into touch with weather agents, is a key element influencing weathering. In rare cases, plants, soil, ice, and other materials may cover a rock, reducing the surface exposed to weather activity.

When compared to rocks whose surfaces are completely exposed to weather, such rocks weather slowly. Weathering is also affected by the slope of the terrain. When the slopes are steep, loose materials are moved downhill by gravity or erosion, resulting in constant exposure of fresh rock.

iv) Mineral composition: The chemical characteristics of a rock are heavily influenced by the mineral makeup. A mineral in a rock can easily react with acids, water, or oxygen, causing significant weathering. Even weakly acidic rainwater, for example, can have a significant impact on limestone. Granite, on the other hand, which is largely composed of silica, is unaffected by such chemicals.

Physical weathering is also affected by mineral composition. Soft rocks are more easily abraded than hard rocks. The solid crystalline rocks are highly compact, with very few openings, and are very resistant to water penetration, thus they can withstand weathering.

v) Time: Weathering takes time. The extent of weathering is determined by the amount of time a rock is exposed to weather. Even the most powerful rocks will be heavily weathered after hundreds of years.

Interplay between these factors determine the extent of weathering and the amount of clays produced as a result of weathering. Lot of studies have been done on the different aspects of clay mineral formation and its application in studies related to weathering.

Sediments produced as a result of weathering is deposited in various depositional environments like flood plain, river bed and ocean floor etc.

(a) Flood plains: When the discharge of water is higher than the capacity of the channel, water flows over the banks and out onto the flood plain where overbank or flood plain deposition occurred. Mostly suspended loads are silt and clay size debris is found. If the flow is high, sufficient to take sandy debris in suspension and subsequent drop in velocity helps the deposition of the silty and sandy suspension. Silt and sand deposited over the flood plain and sediments as a thin sheet are (a) very thin and thin beds normally graded from sand to mud. (b) Prove of initial rapid flow and formation of soil. (Nichols, 2008)

(b) River beds: (i) Sediments in a stream and waterway are the characteristic part. Sediments travel downstream in suspension when water speed is high or turbulent. At the point when there is decrease in speed particularly deeper area zones waterway silt will settle down and the fine material kept. (ii) In stream and river sediment exist in two structures either as suspended load or stored material. It is a fine sediments i.e mud and sediment that stored. Under high stream sand may enter the suspended load and under low stream sediment may settle down the stream bed.

(c) Ocean floor sediments: Generally, these types of sediments are three types

(i) Terrigenous sediments: -Terrigenous sediment are originated from the upper land region from arrive. Aground, rocks are isolated by physical and compound weathering frames. Physical weathering breaks rocks into pieces stretching out from massive rocks to soil. Weathering helps to disintegrate of the source rocks and built up terrigenous sediments. A bit of the rock material is really split down away, which is the source of broken-down particles in seawater.

(ii) Pelagic sediment: - It is the mixture of the clay particles and the micro skeletons of marine organisms of biogenic ooze that is consistently rich in foraminifera with 60% pelagic and neritic grains. It is present in red mud, with under 40% siliciclastic and volcano clastic grains. It can moreover be a silica flood Pelagic deposit are put away at such low rates that they tend to be close near shore. Thus, pelagic sediments are commonly associated with sea floor.

(iii) Hydrogenous sediments: -It is richer in manganese nodules, which precipitate from ocean water and accumulated on the sea depths. Run-off is the major controlling variable for silt deposition. In organic and specific bacterial movement, hydrothermalism process and yearly hydrogenous silt deposited and settled down on the seabed.

Chemical weathering process influences global climate changing and atmospheric CO₂ balance , silicate weathering in particular (Krishnaswami and Singh, 2005) . Major ion measurements in the headwaters of the Krishna and Bhima rivers, their tributaries, and the west flowing streams of the Western Ghats have been used to calculate the rates of chemical and silicate weathering of the Deccan Trap basalts in India (Das, et al, 2004). Only 30% of the Ca in these rivers may have a nonsilicate origin, with basalt and sediment carbonates being a possible source, according to a comparison of estimated and observed silicate-Ca.

The mineralogy and geochemistry of river sediments depends on the lithology of the catchment area, topography, climatic condition, transportation and hydrodynamics of the depositional site (Sensarma et al., 2008). River draining a single lithology helps to study above parameter of the catchment area. Morang river draining Rajmahal basalt was selected as it drains single basalt lithology and there is no report or literature available from this catchment to understand the weathering process.

Different types of weathering are active in a river catchment area and depending on types, the sediment production will be different.

For this project work various published literatures were reviewed to understand the background information and related works. Which helped to formulate research plan for this study.

1.5 REVIEW OF LITERATURE

Previous studies show that Numerous physico-chemical elements, such as rock type, climate, terrain, and exposure age, have an impact on the weathering conditions (Deepthy and Balakrishnan , 2005). Conversely, 1:1 clays (kaolinite) and Fe-Al oxides (gibbsite/goethite) are the most common clay minerals in weathering profiles on the west coast. 2:1 clay minerals are less common. In the rain shadow zone, the clay mineralogy and weathering properties are more complex. There are mainly 2:1 clays and kaolinite. Smectites to kaolinite's transition suggests a humid paleoclimate. Carbonate weathering (Ca Mg) and HCO_3^- , which combined account for around 80% of the chemistry, are the main causes of the cations and anions in the Highland Rivers. The presence of HCO_3^- , excess of (Ca+Mg), and a relatively large contribution of (Na+K) to total cations in lowland rivers (the Chambal, Betwa, and Ken) suggest that silicate weathering and/or contributions from alkaline/saline soils and groundwater may be important sources of major ions for these waters (Sarin et al., 2007).

Climate has an important influence in rock weathering, and rocks and sediments keep a rich record of climate change (Rashid et al., 2014). Interstratified mica-vermiculite forms at the cost of mica in one peaty podzol and is the dominating mineral in the Eh horizon. Interlayering in the interstratified phases appeared to be pH-dependent, with implications for soil and freshwater acidification (Bain et al., 1990). More extensive lateritic weathering than that seen around the mouths of bigger rivers is suggested by the increased abundance of kaolinite and gibbsite in deep-sea sediments near small tropical South American rivers.

According to Millot (1953), in Sudan there exists Niger River that contains large amounts of sediments in which mineral kaolinite, that is common in the nearby

lateritic soils. According to Claridge (1960), the most easily erodible rocks in the watershed are the argillites and bentonitic mudstones that make up the Waipaoa river catchment in New Zealand. These rocks are also abundantly reworked into smectite. Illite, chlorite, kaolinite, and pyrophyllite, all minerals found in the local crystal rocks and Cenozoic soils, were discovered by Lafond (1961) in the Vilaine River in West France. Müller (1961) discovered that the Quaternary alluvium Hadramant, South Arabia, received its supply of fibrous clays from early Tertiary limestones, and that palygorskite (attapulgitic) appeared to go further downstream than sepiolite. Detrital palygorskite was also found by Ravi and Sys (1967) in the Tigris river's flood plain, which had less smectitic minerals than the Euphrates. According to Packham et al. (1961), the composition and character of the nearby argillaceous formations were mirrored in the clay assemblages from nine English river deposits. The three major rivers that enter the north eastern Gulf of Mexico carry clay suites that are made up of both parent rocks and local soils (Griffin 1962). Since the Mobile River is situated in a transitional area, it offers the sea an intermediate clay mineral suite while the more weathered Apalachicola River gets weathering products mostly produced by kaolinite soils. The average composition of the predicted sea sediments is smectite-rich because the Mississippi suspended load is so significant relative to that of two other rivers. The abundance of illite in the upper sub-basin, the northwestern half of the middle basin, and small portions of the lower sub-basin results from a cool, subhumid climate acting on sloped plutonic and metamorphic rocks; the remainder of the Columbia basin is mostly made up of basaltic plateaux undergoing an arid to semi-arid climate, which favours the growth of thorite (Knebel et al. 1968). The sediment mineralogy at the mouths of 25 rivers in Provence, Southeast France, is studied by Charley and Picard in 1970. The mixed-layer minerals and species with poor crystallisation found in the river clays constitute a diversity of rocks. Among others, illite, smectite, apatite, and augite may be found in Estérel, calcareous Provence, Maures, Provence, and the Nice region. In general, metamorphic (Maures) and volcanic (Estérel) regions are more reliable for thick sand minerals. On the other hand, clay minerals are particularly useful sources of indicators in sedimentary contexts. The average size of the many species susceptible to hydrodynamic pressures is the main factor controlling the distribution of minerals in suspended debris, as demonstrated in the Amazon and the Mississippi. The majority of the fractions are made up of nonclay minerals, whereas smaller fractions gradually

record increasing proportions of smectite, chlorite, illite, and kaolinite. Due to their form buoyancy and electrochemical characteristics, tiny kaolinites frequently settle much more quickly than small smectites. Illite is especially prevalent in high-latitude rivers (Mackenzie, St. Lawrence), as well as areas draining young mountain ranges (Indus, Ganges, Brahmaputra), or desert regions (Orange). Kaolinite dominates a significant portion of tropical rivers, although chlorite is frequently uncommon or absent as reported from Niger, , Orinoco, Parana nd Amzon river (Gibbs 1977). Smectite has been found in tropical and subtropical rivers like the Caroni, bulk suspensions. Orange, Niger, and Nile While calcite and dolomite are typically found in rivers draining either dry or hilly environments, quartz and feldspars are frequently seen.

Volcanic rocks often experience more rapid weathering than the majority of other rocks. Porous rocks, such as pyroclastites, encourages leaching and drainage, whereas the prevalence of amorphous minerals aids hydrolysis. Andosols are a word used to describe soils that develop on volcanic rocks quickly and become fairly thick. Some minerals that are typically generated at low latitudes or altitudes are able to develop under less hydrolysing circumstances due to the quickness of the volcanic rock weathering, which frequently results in an azonal distribution of soil clays. Smectites, which readily form on Fe-Mg species and on thiolitic materials more frequently as Al species, are a prime example of this. Smectites, which are abundantly formed on volcanic rocks in temperate-warm to chilly climates, are moderately to extremely strongly crystalline. In hot to moderate climate zones, analogous alteration leads to the abundance of kaolinitic minerals. According to the drainage, climate, petrography, and amount of weathering, four stages of the development of pedogenic clay may be schematically distinguished.

1. Amorphous to subamorphous clays, such as imogolite and allophane, are formed during the initial weathering stage. Frequently described in Japanese andosols (Yoshinaga and Aomine 1962; Henmi and Wada 1976; also see the syntheses of Wada 1977; Sudo and Shimoda 1978); imogolite also occurs in surficial weathering crusts on basalts of the Massif Central, France (Torrent et al. 1982). Clays that are not crystalline appear to be more frequently maintained in immature andosols and to have a limited life span (Wada and Aomine 1973, Wada 1977).

2. Under moderate circumstances, the next stage is characterised by the formation of Fe- to Al smectites (nontronite, montmorillonite, etc.). In Europe, volcanogenic smectites have been reported from the Mediterranean to Iceland, it make up the majority of soil profiles (Hétier et al., 1977). Organo mineral complexes may develop in organic-rich andosols and stop the swelling-collapse effects often seen by successively glycolated and heated smectite; as a result, smectite behaves like swelling chlorite or pseudochlorite. (Moinereau 1977).
3. The expansion of halloysite, a hydrated kaolinite, is what distinguishes the third phase. Within the tropical soils exposed to heavy rainfall and a brief but considerable dry season, halloysite constitutes a very stable mineral. According to Her billon et al., this is the situation with young soils produced on basalts in Cameroon (Sieffermann et al., 1968). (1981) In Burundi, combined layers of kaolinite and smectite are made up of halloysite or metahalloysite, a transitional phase where smectite alters to form kaolinite.
4. Extended kaolinization, which is frequently related to the development of gibbsite. In humid intertropical conditions, where the effect of parent rocks is only secondary to that of the climate and leaching, kaolinite and gibbsite easily form. Halloysite is replaced with kaolinite, which is particularly common where there is strong drainage and no discernible dry season (e.g., Sieffermann et al. 1968, Chamley 1969, Kantor and Schwertmann 1974 Quantin et al. 1975). According to Rutherford and Watanabe (1966), in Papua-New Guinea, gibbsite is more likely to occur when weathering complexes are more advanced and volcanic eruptions are older.

Mainly three mechanisms are responsible for the formation of clay minerals; Inheritance, Neoformation and Transformation (Ref). Clay minerals can be used to understand the sediment production in a catchment area, weathering process, subsequent transportation and related surficial processes. Rivers have an important role in the evolution of Earth's surface and for transporting sediment from land to sea.

Clay minerals are used to determine the age of suspended sediments, weathering severity, and source area. The Madeira and Maraón–Solimes Rivers, Smectite

concentrations rise as it moves downstream. kaolinite content of all shield tributaries is high, lateral bank erosion also plays a role, as more mature sediments high in smectite are introduced into the river. Clay mineral abundance has also been depending on the sediment type, montmorillonite up to 67 percent; and kaolinite are found in the Godavari and Krishna assemblages.

Depending on the species; clay crystals are primarily composed of aluminium or magnesium, oxygen, silicon and hydroxyl (OH), along with a variety of related cations. There are two types of sheets,

(1) The tetrahedral sheets are often composed of T.O, where T is a tetrahedral cation that is mostly Si with various amounts of Al or Fe. The four corners of the tetrahedron are formed by oxygen anions, while silicon occupies the middle. The individual tetrahedra form a hexagonal mesh pattern when they are joined with nearby the three basal oxygens as a tetrahedral form. The fourth tetrahedral corner indicates the orientation normal to the sheet. The octahedral sheet of it includes the apical oxygen (Drits et al.,1997).

(2) Octahedral sheets, Medium-sized cations (typically Al, Mg, Fe, or Fe) are located in the middle of the octahedral sheets, while oxygens are located at each of the eight corners. By exchanging oxygen, the individual octahedra are linked laterally with the neighbouring octahedra and vertically with the tetrahedra. Three octahedra make up the smallest structural unit of the octahedral sheet. Tri-octahedral sheets are those features having octahedral cations (bivalent ions like Mg, Fe²⁺) at the core of all three of its octahedra.

A layer is the structure created when tetrahedral and octahedral sheets are assembled. According to Boettcher (1966), one tetrahedral sheet and one octahedral sheet make up the 1:1 layer, sometimes referred to as the T.O. layer. This layer is typical of the kaolinite group and consists of an unshared plane of anions in the octahedra of the sheet of groups. The 2:1 layer, sometimes referred to as the T.O.T. layer, joins two external tetrahedral sheets with an octahedral sheet. Since the relative arrangement of both tetrahedral sheets is inverted, all apical oxygens point in the direction of the tetrahedral sheet and can be shared. The majority of clay species belong to the 2:1 layer type, including smectite (Drits et al., 2002).

The basal spacing between two successive units is 7 Å° for the 1:1 type, 10 Å° for the 2:1 type, and 14 Å° for the 2:1:1 type. A clay particle, which is typically less than a few micrometres, is created when a large number of structural units are piled together (um). Ionic substitution is the main driver of several distinctive properties of clay minerals, such as adsorption-desorption, expandability-retraction, and the formation of organo-mineral complexes.

Kaolin-Serpentine Group

Minerals of the 1:1. (T.O.) clay group can be either di-octahedral or tri-octahedral. The following are the major species:

Kaolinite $\text{Al}_2\text{SiO}_5(\text{OH})_4$: With XRD, preliminary identification may be made by observing the fundamental (001) basal reflection of kaolinite at 7.14 Å° with higher orders at 3.57 Å° (002) and 2.38 Å° (003). As with the other clay minerals, these basal reflections are most easily observed by preparing an aggregate with preferred orientation, often by simply drying down a suspension onto a glass slide. Confusion with other clay mineral groups, such as the serpentine or chlorite minerals, which also yield a strong 7 Å° basal reflection, can usually be avoided by careful measurement of the complete powder pattern in a randomly oriented specimen or by the use of supplementary treatments (Mac Ewan & Wilson 1980). Thus, heating at 600 °C reduces kaolinite to an amorphous state whereas the serpentines and chlorites retain their crystalline structure. Again, kaolinite is generally resistant to acid treatment, such as 6 M HCl at 95 °C for 30 min. whereas chlorites and serpentines usually decompose completely. Finally, chlorites and serpentines do not form intercalation complexes in the same way as kaolinite. Kaolinite is a di-octahedral clay mineral that is triclinic and well ordered, and whose crystals frequently exhibit pronounced hexagonal outlines. Kaolinite mostly develops during early diagenesis, but it may also happen in surficial settings through pedogenetic processes. Although nacrite and dickite are both monoclinic and almost orthorhombic, respectively, they often belong to distinct hydrothermal settings and have big crystals. Halloysite is a kind of hydrated kaolinite that frequently has a tubular or glomerular structure.

Mica Group

Mica structure composed of strong negative charges offset by substantial univalent interlayer cations and apparent replacements in the 2:1 layers. The charge results: (1) from substitution of trivalent cations (R^{3+}) for Si^{4+} in tetrahedral positions, (2) from substitution of univalent (R^{1+}) or bivalent (R^{2+}) for R^{2+} or R^{3+} in octahedral positions, or (3) from vacancies in octahedral positions (Bailey 1980a). The majority of the cations found in interlayers are univalent and linked to either no or very few water molecules. This large group exhibits well-formed but erratic edges on electron micrographs and a steady spacing on X-ray diagrams. It consists of both dioctahedral and tri-octahedral shapes (Wilson et al. 1984).

Mixed-layers

i) Different type of stacking sequence of single layers defines mixed-layered or interstratified clays, which are noteworthy phyllosilicate formations. The involved layers might be of the 2:1, 2:1:1, or even 1:1 sorts. The potential of interstratification or mixed layering arises from the relatively weak chemical and structural ties that connect the various layers of a certain clay particle. Because all different types of clay layers exhibit sheets of oxygen and hydroxyl that are arranged in a hexagonal mesh on their basal surfaces, layers with different internal layouts can stack together and yet have moveable interfaces. Mixed-layered clay minerals are frequently found because of their structural characteristics, which are virtually entirely unique in the range of mineralogical traits (Weaver 1956, Reynolds 1980). The majority of mixed layers are really clay species that are in between two separate clay minerals. Clay minerals developed in weathering, diagenesis, or hydrothermal and sedimentary settings. Interstratification may be divided into three categories. A periodic alternation of layers of two types A and B refers to regular or ordered mixed-layers ABABAB or AABAABAAB (Lucas 1962).

2. Clays with uneven and randomly mixed layers correspond to a random alternation of each type of layer. This group frequently correlates to X-ray diffraction patterns with dome, plateau, or wedge-like forms and locations that fall in between those of the individual minerals involved (Lucas et al. 1959, Millot 1964, Cradwick and Wilson 1978).

3. There seem to be somewhat organised structures, notably in certain soils (Mac Evan 1949), although they are not well recognised and are not given a formal name.

The reasoning and interpretation regarding mixed-layer clays may completely diverge if these minerals are identified by X-ray reflections that are located between those of single clay minerals (for example, Lucas 1962, Holtzapffel 1985), or by statistical calculations that take into account the positions of the single minerals themselves (Reynolds 1981 Srodon 1984a, Srodon and Eberl, 1984).

Clays are particularly challenging to precisely characterise because the minerals are sometimes very tiny, inconsistently or weakly crystalline, and linked to sub-amorphous compounds. Transitional words and complicated linkages are commonly used. There are presently no accepted preparation, identification, or quantification techniques (Pierce and Siegel 1969, Brindley and Brown 1980, Stucki and Banwart 1980, Mann and Fischer 1982, Thorez 1983).

Weathered materials from land to the ocean are transported by rivers. There is large production of elements from both natural and anthropogenic sources; from this we can determine the chemistry of river waters. Iron-rich illites sometimes occur in microcrystalline, sand size, peloidal aggregates and may be grassy green in colour, non-pleochroic and display low-order yellow birefringence (Baker 1997). Trioctahedral illite may show pleochroism in brown to yellowish brown colours (Ushizawa 1981) and where it derives from the weathering of biotite, some of the optical properties of the oxidized parent mineral (Fordham 1990b). The formation of illite has been described in a variety of environments spanning a wide range of temperature and pressure conditions. In general, however, it seems that formation at earth surface conditions is somewhat more controversial than formation following deep burial, low-grade metamorphism or hydrothermal activity. I/S, in all its variants of random, partial and long-range order, forms through a range of diagenetic to very low-grade metamorphic environments and through hydrothermal alteration. Less certainly, I/S is also thought to form in weathering environments. Here, selected accounts of I/S formation in these different environments are reviewed with particular emphasis on the mechanisms of formation involved. As indicated in the work of Walker (1957 1958, 1975), however, as well as that of Brindley (1966), the response to ethylene glycol may not provide an unambiguous means of differentiating between smectites and vermiculites (see also MacEwan & Wilson 1980). Thus, Brindley (1966) found that many types of vermiculite, even high-charge varieties, did swell to

a considerable extent with ethylene glycol, irrespective of the interlayer cation. With glycerol, the results were more consistent.

1.6 RESEARCH PROBLEM

Sediments associated with Rivers include bed load and suspended load. Mineralogical study of bed load sediments will help to understand the weathering process, the intensity of weathering and the denudation in various environmental conditions, origin, mode of occurrence, biological, physical and chemical processes, mobilization and transport of various elements. River draining a single lithology helps to study above parameter of the catchment area. So here, Morang river draining Rajmahal basalt was selected as there is no report or literature available from this catchment to understand the weathering process.

1.7 OBJECTIVE

The objectives of this study are:

- a) To understand the mineral composition of Morang river bed load
- b) To understand the process of clay mineral development in the catchment region.

CHAPTER-2

STUDY AREA

2.1. Study Area

The study area comes under the Sahebganj district in the N-E part of Jharkhand (India), surrounded by a small portion of the district of Bhagalpur in the north, Kathiari district of Bihar state in the east, and West Bengal to the south. Unlike Jharkhand, where the districts of Pakur and Godda are to the south and west, respectively. The study area is situated between 24°50'52.08" and 25°10'59.8"N latitude and 87°33'16.16" and 87°42'36.8" E Longitude.

2.2 Climate and Geomorphology

(a) This region receives rain from the south-west monsoon. Period between June and September marks the rainy season. In the area, 1575 mm of rain falls on average annually.

(b) This region has a sub humid to barely humid environment. The weather is warm from March until mid-June. From mid-June until the end of September is the rainy season. Winter begins in mid-November and lasts through the end of February. summer's heat from March through May. Minimum temperature is 6.8°C, and the maximum temperature is 44.4°C.

(c) The Ganges, Gumari, and Morang rivers mostly drain the Sahibganj district's geomorphology. All of these rivers, with the exception of the Ganges River, are seasonal. The Gumari and Morang Rivers contribute significantly to the primary surface run-off during the monsoon. The district's principal geomorphological characteristics include a northern scarp, flat alluvial terrain along its eastern edge, and the resilient lava plateau of Rajmahal, which makes up the majority of the district.

(d) The Rajmahal type soil, which is generated from basaltic lava, is the main soil type present in the Sahibganj district. Other soil types that may be found in this area are red soil, alluvial soil, eroded scarp soil, tal soil, and foothill soils. Alluvial soil of the clayey loam type may also be found close to the Sahibganj plains.

(e) The structure of the Morang River basins is leaf shaped. These structures is stretched along the main river course and open in broad like structure and lower part is compressed and opens out like narrow band and mixed with Ganga river.

2.3 Lithology and Lithostratigraphy

Morang river basin composed of two group or formations. Rajmahal formation having thickness 600m and Dubrajpur formation having thickness 122 to 137m. The lithology of this area is coarse sandstone, shale, clay beds, pebbly sandstone, fossiliferous siltstone, baked sandstone, siltstone, laterite, soil alluvium etc. (Mahadevan, 2002).

1. Dubrajpur formation: - It is situated between the underlying Barakar and the overlying Rajmahal trap. It contains conglomerate, ferruginous sandstone, coarse pink felspathic sandstone, red and brown siltstone and clay. It occurs as narrow discontinuous bands striking the Rajmahal formation along the western margin of the Rajmahal hills. Three major exposure of Dubrajpur formation by Sengupta (1985) are as follows:

- (i) The saravanpahar located 3.35km ENE of Nargraj.
- (ii) Section of Katangi hill, 380m south of Kharikasol.
- (iii) Chaparia pahar section, 4.8km north of Narganj.

2. Rajmahal formation: It is placed above the Dubrajpur formation and it is in contact with the Barakar onto the crystalline basement along the western margin. It has maximum thickness of 330 m and thins 100 m towards near the eastern of the Raniganj basin. The beds of the sandstone, carbonaceous shale, grey shale and tuff are mainly found in this Rajmahal area. The Rajmahal formation may be divided into four major successive zones such as:

- a) Basalt lower trap flows with unfossiliferous in tertrappean beds
- b) Successive lava flows with highly fossilifeorus in tertrappean beds
- c) Overlying lava flows with scarcely fossilifeorus in tertrappean bed
- d) Lava flows with no inter trappean beds

Table 1 Lithostatigraphy of Rajmahal formations (simplified from Sengupta,1988, Mahadevan,2002)

FORMATION	LITHOLOGY	AGE
Rajmahal formation	Soil, Alluvium	
	Laterite	
	D. Upper trap flows with and without intertrappean scarcely fossiliferous	Cretaceous
	C. Upper trap flows with fossiliferous intertrappean bed 8 with Taeniopteris spatulata-Brachyphyllum rhombicum. Assemblage zone 3	Lower cretaceous
	B. Lower trap flows(4 to 6) with fossiliferous intertrappean beds(4 to 7) with Cladophlebis indica-Dictyozamites indicus assemblage zone 2	Middle Jurassic
	A. Lower trap flows (1 to 3)with unfossiliferous intertrappean beds(1 to 2)	
Dubrajpur formation	IGNEOUS CONTACT	Lower Jurassic
	Baked sandstone, siltstone	
	Fossiliferous siltstone with Ptylophyllum acutifolium and Gleichenites gleichenoides Assembly zone-1	
	Pebbly sandstone	
	Coarse sandstone, shale and clay beds	Triassic

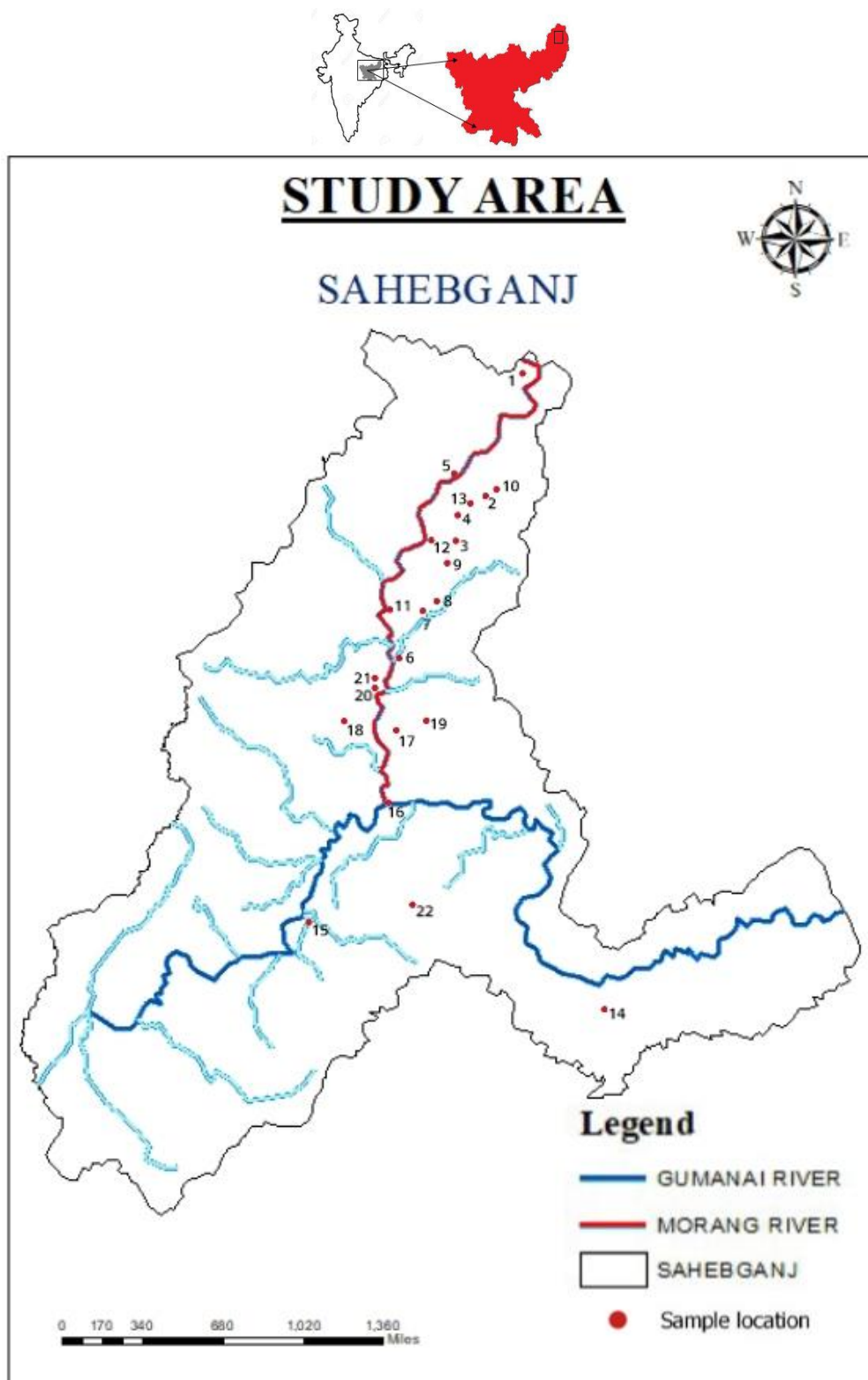


Figure 1: Drainage basin of Morang (in red) and Gumani river (in blue) and sampling location marked as red dot with number. (source – Bhuvan portal watershed)

{Comment: Check the study area location. Is it only Sahibganj district or spread over other district.
No place mark on the map. No lat and long information}

CHAPTER 3

MATERIAL AND METHODS

3.1. Sampling

Total 21 river sediments samples were collected by Haldhar kumar and Premchand Kisku (PhD scholars, Department of Geology, Central University of Punjab) from different regions of the Morang River catchment. The total length of the sample collected area is about 128.38 km. The samples are collected from inside the catchment area of the river. All the information related to sampling area, like latitude, longitude, altitude sample collected area and their sample location distance are described in Table 2.

Table 2: Sample number and location of the river bed sediments.

Sample no.	Latitude	Longitude	Altitude In(m)	Sample Location name
Loc1	25°10'59.8"	87°40'1.3"	149	Durgatola
Loc2	25°07'07"	87°38'51.2"	90	Sabaiya
Loc3	25°05.701'	87°37.911'	81	Dighi
Loc4	25°06.508'	87°37.974'	82	Turua
Loc5	25°07.8223'	87°37.862'	89	Kordhia
Loc6	25°01.9858'	87°36.124'	67	Chapgawan
Loc7	25°03.483'	87°36.87'	72	Simra
Loc8	25°03.787'	87°37.312'	73	Gorda
Loc9	25°0459.6"	87°37'38.4"	77	Majhauna
Loc10	25°07.329'	87°39.198'	104	Bhurkunda
Loc11	25°03'31.25"	87°35'49.98"	72	Chhotakusma
Loc12	25°05'43.2"	87°37'08.30"	82	Matia
Loc13	25°06'52.99"	87°38'22.48"	85	Sabaiya
Loc14	24°50'52.08"	87°42'36.8"	38	Amjhor
Loc15	24°53'37.67"	87°33'16.16"	56	Simaldhab
Loc16	24°57'24.64"	87°35'46.82"	60	Phuibhanga
Loc17	24°59'42.14"	87°36'1.93"	54	Bara beyasi
Loc18	24°59'59.6"	87°34'23.36"	64	Chasganwan
Loc19	25°00'01'0.3"	87°36'59"	76	Rampur
Loc20	25°1'2.40"	87°35'21.40"	69	Bansjori
Loc21	25°01'21.3"	87°35'21.4"	67	Kalkha

3.2 Analytical methods:

3.2.1 Sample Processing:

All the 21 samples were dried in hot air oven at 50° temperature until the samples are well dried. Pebble sized fragments, organic materials like plant roots were removed, and the samples were disintegrated with the help of agate pestle and mortar. Using cone and quartering method samples were homogenized to avoid biasness further it was sub-sampled. The detailed procedure of this step is:

- $\leq 2\text{mm}$ sediment samples are mixed and poured into a conical pile by the help of butter sheet.
- Sample cone was divided into 4 quarters.
- From this, two diagonally opposite quarters were taken for further analysis and the remaining sample was kept as stock.
- Weight of the samples were recorded at each step (Table 3)

Table 3: weight (g) of different fractions of river bed sediments

Sample Name	Wt. of dried sample (g)	Wt. of coarser sediments (g)	Wt. of <2mm size sediment (g)	Weight of sample taken for clay separation (g)
Loc1	94	10	84	12
Loc2	189	108	81	10
Loc3	156	62	94	10
Loc4	133	-	133	10
Loc5	192	121	71	10
Loc6	78	-	78	11
Loc7	210	210	-	-
Loc8	246	136	110	10
Loc9	94	-	94	10
Loc10	145	67	78	10
Loc11	178	59	119	11
Loc12	251	213	38	10
Loc13	98	-	98	10
Loc14	159	-	159	11
Loc15	132	27	105	10
Loc16	81	8	73	10
Loc17	78	78	-	11
Loc18	126	30	96	11
Loc19	350	162	188	12
Loc20	161	108	53	11
Loc21	211	178	33	10

Further, the samples were powdered using agate disk mill (Retsch 200). Then 3– 5g of fine sediments was sent for XRD and analysis.

3.2.2 Sample analysis:

Clay minerals are produced by weathering process either sub aerial or sub-aquatic condition. Clay minerals are important for paleoclimatic studies. The clay mineral assemblage can be correlated with climate and rock type (Deepthy and Balakrishnan 2005). They exhibit less than $2\mu\text{m}$ size. Clay mineral characterization was carried out by X-ray Diffractometry (XRD) technique.

3.2.2.1 Separation of clay minerals from bulk samples

Samples were mixed with EliX water and kept in ultrasonicator for 30 minutes to disaggregate sediments. Organic matters were removed by adding dilute H_2O_2 (Figure 2) and kept for overnight (Deepthy, 2008).



Figure 2: Photographs of sample slurry mixed with hydrogen peroxide



Figure 3: Photographs of samples kept in the Measuring cylinder

After this treatment, sample slurry was transferred into 1000 ml measuring cylinder and volume was made up to 1000 ml (Figure 3). According to Stokes law, the clay fraction was separated. The sample mixed with EliX water was kept for 3h 36 min after shaking. Later The top 5cm was siphoned out (Figure 4). Here the time of

withdrawal of clays was calculated from Stokes' law. This procedure was repeated thrice to get enough clay. Stokes' law for velocity (v) of the clay particle is

$$V = \frac{2ga^2(d_1 - d_2)}{9\mu} \quad \text{where, } g = \text{acceleration due to gravity}$$

9μ

a = radius of the particle (taken as sphere)

d_1 = density of the particle, d_2 = density of the medium

μ = viscosity



Figure 4: Photographs taken during separation of clay from sample

Separated clay fraction was concentrated by centrifugation method. The centrifuge tube was filled with sufficient amount of sample in suspension. At suitable RPM, clay size fraction was separated in tube after centrifugation. Further samples were made into different aliquot for different treatment. Two different aliquot about 3 ml each was taken in a centrifuge tube for addition of 1 M potassium chloride and calcium chloride separately. Mixture was thoroughly mixed using a cyclo-mixer. This was repeated four times. Following this, the sample was rinsed with distilled water, then washed with 50% and 100% alcohol, respectively. Cyclo-mixer was used prior after each step of washing and centrifugation. Using micropipettes, samples were placed on top of glass slide dropwise (Figure 5) further, it was air-dried. Once samples were dried completely, it was taken for XRD analysis.



Figure 5: Photographs of samples placed on glass slide. Oriented clay slides were prepared for XRD analysis



Figure 6 - Photographs of XRD facility at Central Instrumentation Laboratory, Central University of Punjab.

3.2.2.2 X-ray powder diffraction (XRD) analysis:

Powder XRD is mainly used for identifying the phase of different minerals as well as crystalline materials. Based on atomic spacing and crystal structures different phases are identified. It is a quicker and non-destructive method of analysis for mineral phase. Working principle of XRD consist of following components such as: X-rays produced from cathod ray tube, these rays are filtered to produce the monochromatic radiation, finally collimated and directed towards the samples. Further, a detector detects the diffracted X-ray signal from sample electronically or by microprocessor, which converts the signal to count rate. This principle is on the

basis of Bragg's law ($2d \sin\theta = n \lambda$ where n is the integer, λ is the wavelength of X-ray, d is spacing between the interplanar spacing between the crystal and θ is the angle of incidence X-rays beam (Flohr, 1997)). X-rays are electromagnetic waves up on hitting the crystals having regular arrangements of atoms, it interact with electrons of atom to scatter the X-rays as elastic scattering. A regular array of spherical waves results from a regular array of scatters interfere destructively with one another in the majority of directions, cancelling each other out. Finally these waves are registered in the detector.

3.2.2.3 XRD spectrum analysis:

XRD data help to identify the variation in mineralogical composition according to their distance of the sample collected locations. The relative intensity percentage (I in %), d-spacing [Å] and (hkl) value of the particular mineral were identified with the help of International centre for Diffraction Data (ICDD) database for each spectrum. River bed sediments collection distance, mineralogical distribution from upper region to lower part of the river mouth can be identified with the help of XRD analysis. For XRD analysis a software known as X'pert high score was used. It is a comprehensive tool for extensive powder pattern analysis.

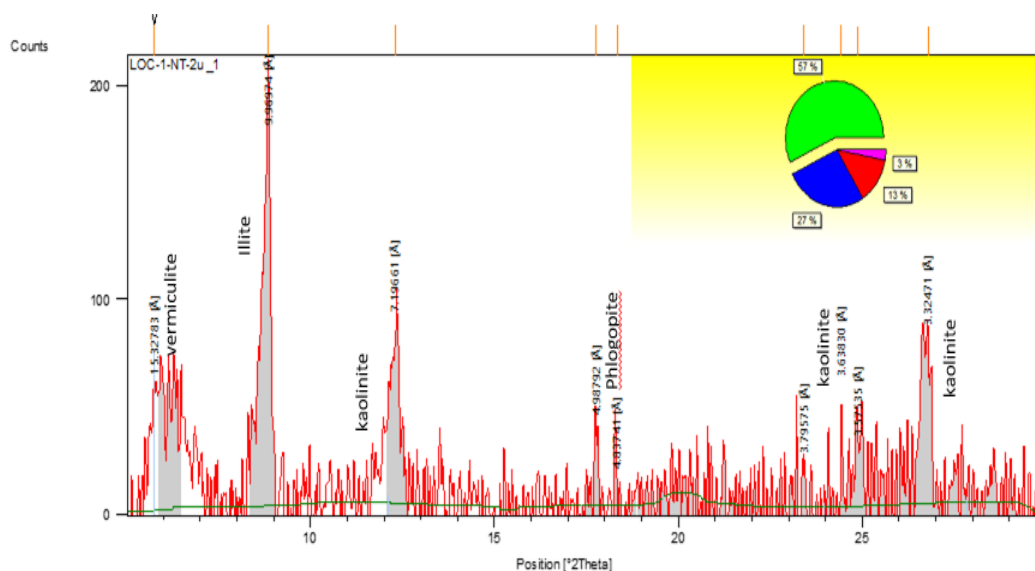


Figure 7 - XRD spectrum of Location 1

Table - 4 Selected clay minerals and its important XRD peak I(%) along with its (hkl) parameters are listed below.

Clay Mineral	I (%)	h	k	l
Kaolinite	100	0	0	1
	51.1	-1	-1	0
	48.1	-1	-1	1
Illite	100	0	2	1
	67	1	1	0
	61	0	0	2
Vermiculite	100	0	0	2
	3.6	-1	1	1
	1.5	0	2	1
Montmorillonite	100	0	0	5
	90	0	0	2
	70	0	0	1
Chlorite	100	0	0	2
	84.1	0	0	3
	63.9	0	0	4

For all typical powder diffraction applications, it combines phase identification, crystallographic analysis, cluster analysis, and Rietveld calculations in a single software package. Based on Rietveld method the important peak listed in the table 4, were identified and semi-quantified using their relative intensity ratio (RIR) value reported in the ICDD library. Semi-quantification, distribution of clay in a sample and labelling of clay minerals along the peak in the XRD spectrum can be seen in the figure 7. For the all the samples, both clay and silt fraction, separate analysis were carried out and semi-quantification was used for further analysis and interpretation.

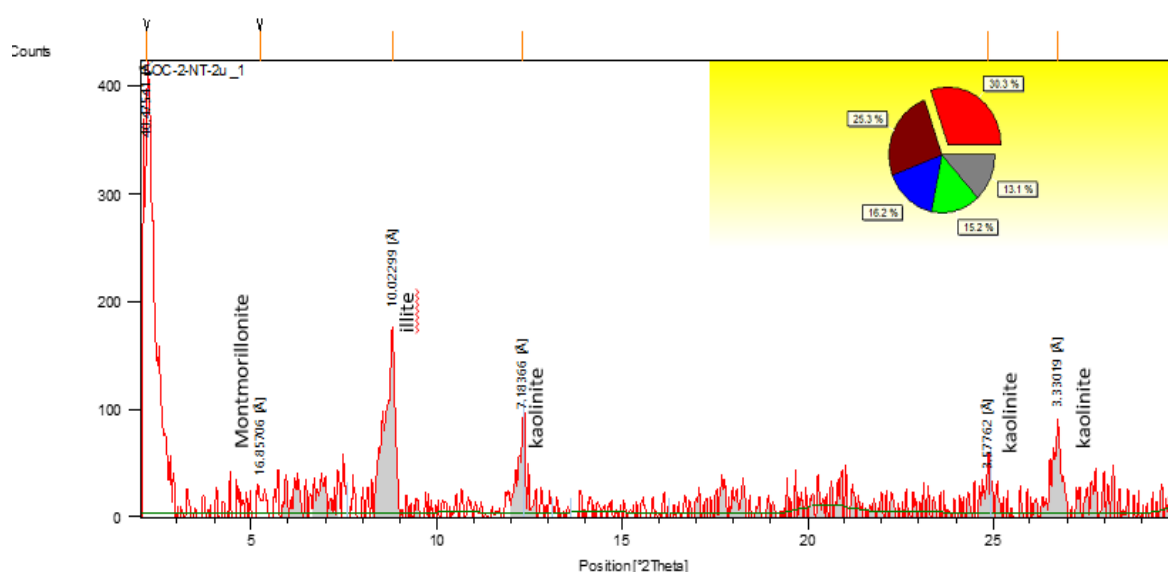


Figure 8 - XRD spectrum of Location 2

CHAPTER-4

RESULTS AND DISCUSSION

4.1 Clay mineral analysis:

Semi-quantification of the clay mineral assemblage from 21 locations (Loc-1 to Loc-21) was carried out using XRD. Percentage of each clay minerals are listed in the table 5. From the result, it is found that all the samples show the presence illite. Except sample LOC12 & 19, all other sample shows kaolinite. Presence of Vermiculite, montmorillonite, clay mica and chlorite are dominated for many samples. The clay mica include clay size fraction of both biotite and muscovite minerals. Comparison of XRD spectrum from all the locations is stacked in the figure 9. Distribution of clay mineral in percentage (semi-quantification) along with other mica-group minerals like muscovite and biotite are compared in the figure 10.

Table 5 – Semi-quantification (in %) of the clay minerals from XRD spectrum for each samples are listed below.

No.	Sample Name	Kaolinite (%)	Illite (%)	Vermiculite (%)	Clay Mica (%)	Montmorillonite (%)	Chlorite (%)
1	LOC 1	27	57	3	13	-	-
2	LOC 2	15	30	13	16	25	-
3	LOC 3	10	28	11	32	19	-
4	LOC 4	19	43	17	-	-	21
5	LOC 5	2	20	20	51	-	7
6	LOC 6	9	33	11	22	16	9
7	LOC 8	15	61	8	-	-	16
8	LOC 9	15	48	19	-	-	18
9	LOC 10	6	30	15	24	13	13
10	LOC 11	6	35	25	-	21	14
11	LOC 12	-	61	-	39	-	-
12	LOC 13	17	42	-	41	-	-
13	LOC 14	13	40	-	47	-	-
14	LOC 15	22	27	16	35	-	-
15	LOC 16	11	89	-	-	-	-
16	LOC 17	34	66	-	-	-	-
17	LOC 18	13	47	-	40	-	-
18	LOC 19	-	42	-	14	44	-
19	LOC 20	20	31	-	48	-	-
20	LOC 21	13	48	14	-	-	-

Counts

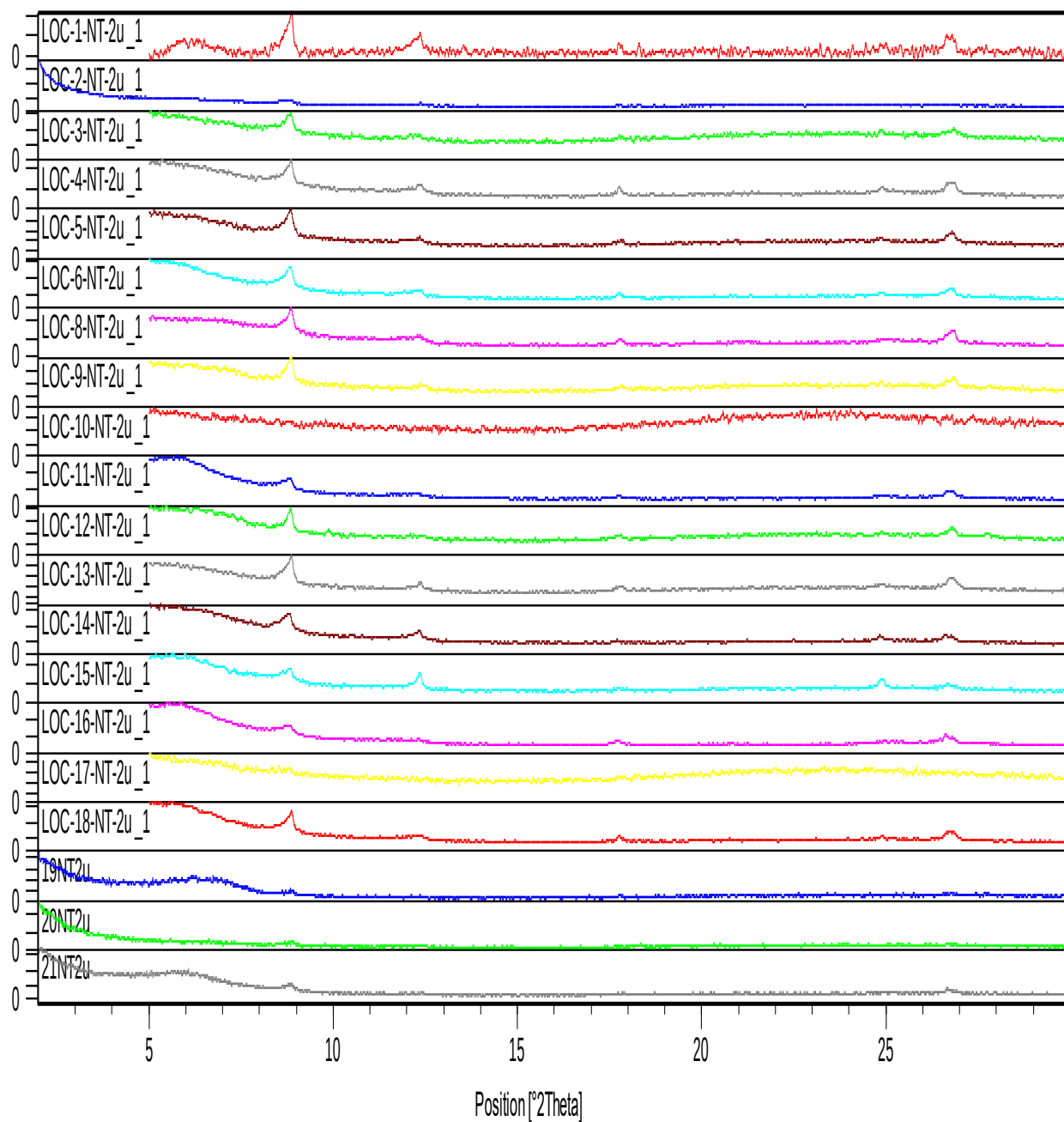


Figure 9 - Comparison of XRD spectrum of clay minerals

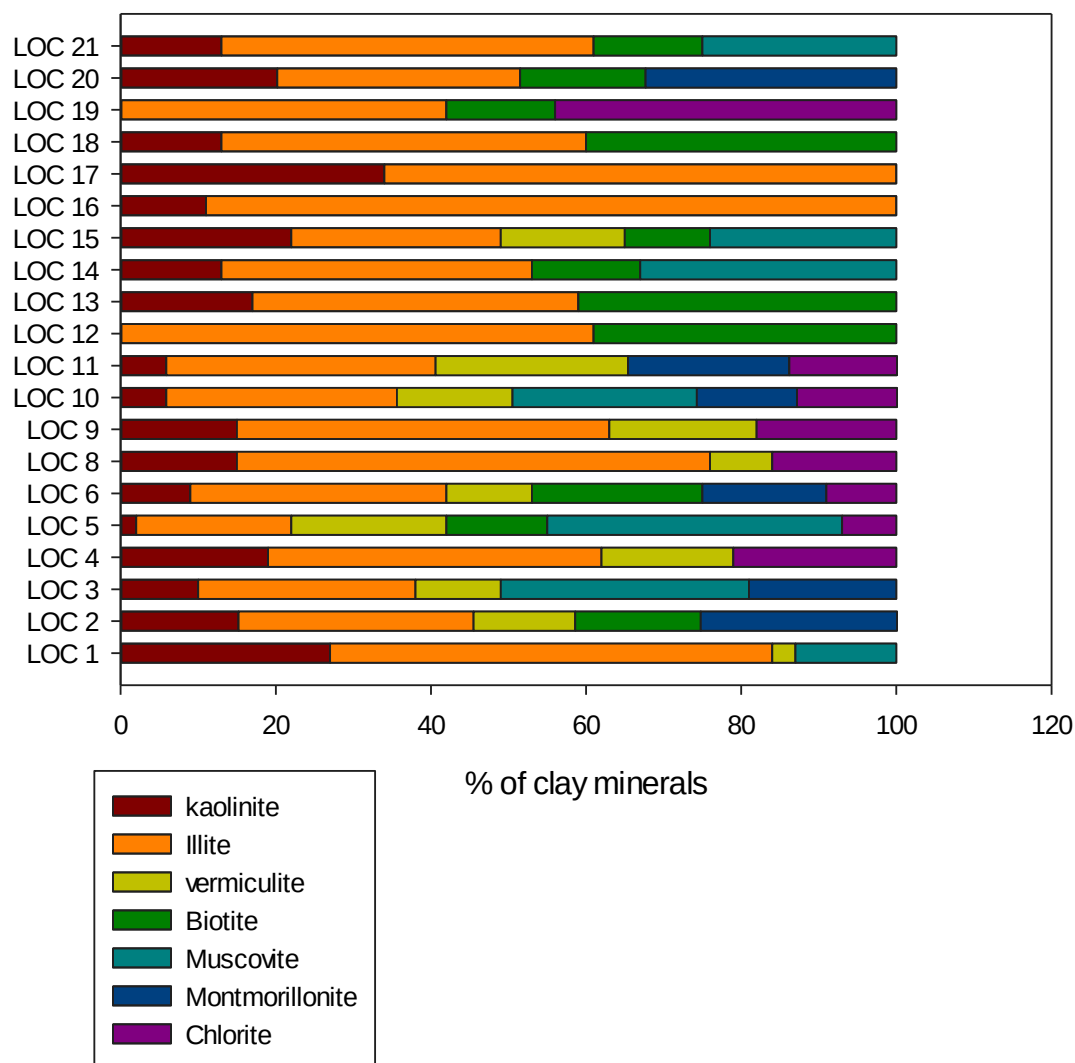


Figure 10 - Semi-quantification of clay mineral percentages the samples.

4.2 Bulk mineral analysis from silt size fraction:

As described in the methodology, silt size fractions were separated by settling method and bulk mineralogical analysis were carried out for each samples. The result shows that the even in the silt size illite percentage is higher compare to other mineral present in the samples. Along with illite, kaolinite is also present in most of the samples. Other clay minerals such as Montmorillonite, vermiculites are also present in few samples at varying percentage. Silt size fraction of feldspar, quartz, enstatite, muscovite & biotite are also found in different samples collected from various localities.

Table 6 – Semi-quantification (in %) of the clay along with other minerals analysed from silt size fraction of all samples are listed below (L – less than 2 %).

Sample Name	Illite (%)	Muscovite (%)	Kaolinite (%)	Feldspar (%)	Quartz (%)	Biotite (%)	Enstatite (%)	Montmorillonite (%)	Vermiculite (%)
LOC 1	35	44	9	12	-	-	-	-	-
LOC 2	79	-	21	-	L	-	-	-	-
LOC3	44	49	7	-	-	-	-	-	-
LOC 4	28	17	-	15	L	21	19	-	-
LOC 5	43	47	5	-	-	-	5	-	-
LOC 6	52	39	10	-	-	-	-	-	-
LOC 8	53	32	7	-	L	8	-	17	-
LOC 9	29	36	8	-	-	-	9	-	-
LOC 10	59	24	-	-	-	-	-	-	-
LOC 11	35	45	5	-	-	-	-	13	17
LOC 12	24	38	-	15	L	-	4	14	-
LOC 13	34	17	6	-	-	19	-	-	12
LOC 14	33	32	8	10	L	12	-	-	15
LOC 15	49	-	7	-	L	16	-	-	18
LOC 16	43	38	6	9	-	-	13	15	-
LOC 17	27	20	15	0	-	-	-	-	9
LOC 18	37	41	5	14	-	-	16	-	-
LOC 19	54	-	9	32	L	-	-	-	-
LOC 20	40	38	19	4	-	-	-	-	-
LOC 21	33	39	5	23	-	-	-	-	-

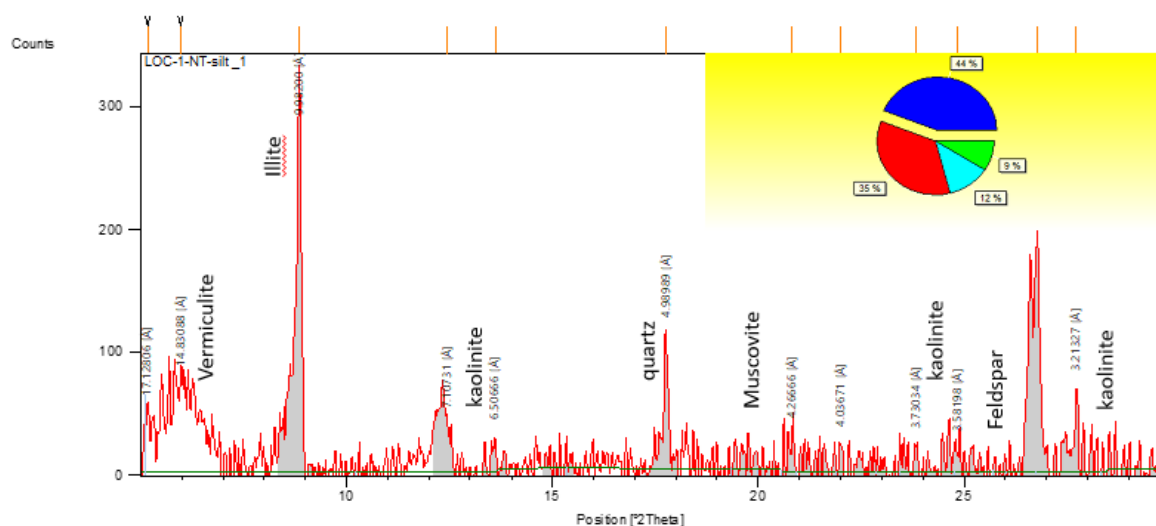


Figure 11 - XRD spectrum of silt size fraction from Location 1

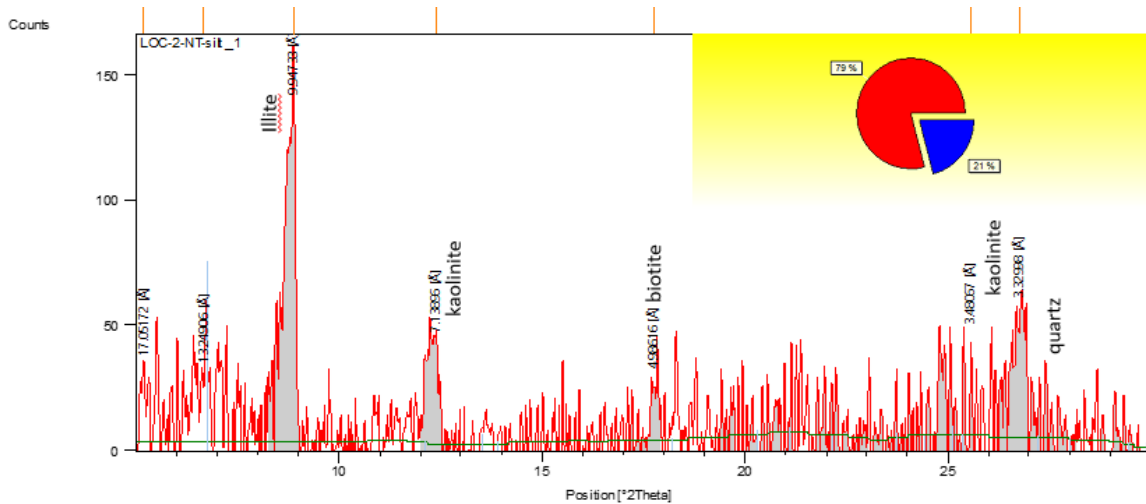


Figure 12 - XRD spectrum of silt size fraction from Location 2

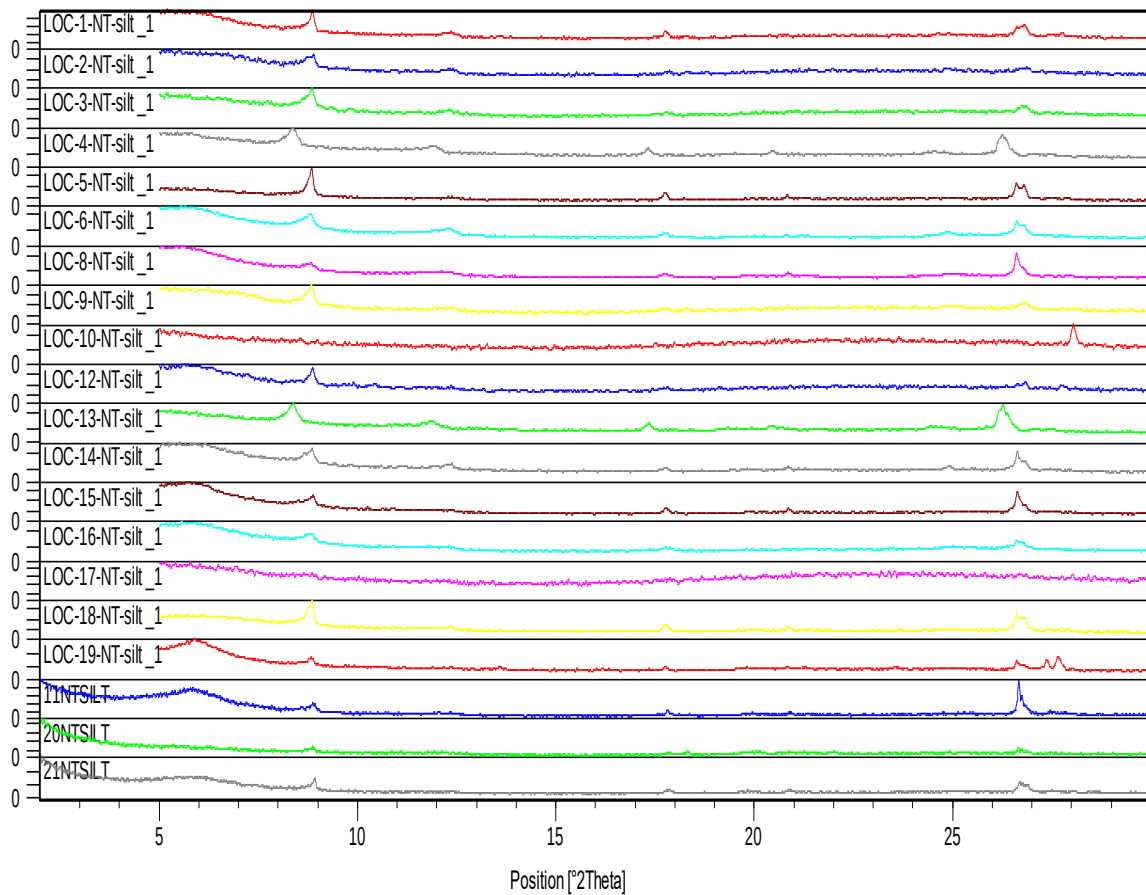


Figure 13 - Comparison of XRD spectrum of silt minerals

Semi-quantification of clay minerals along with others from silt size fraction is listed in the Table 6. Only two representative XRD spectrum of silt size fraction from Location 1 & 2 are given in the figure 11 & 12 to illustrate the peak position, counts and d-spacing along with percentage of minerals in a pie diagram as an insert, in the

sample. Comparison of XRD spectrum of silt size fraction from all the locations is stacked in the figure 13. Distribution of clay mineral in percentage (semi-quantification) along with feldspar, quartz, enstatite, muscovite & biotite are compared in the figure 14. While comparing the clay and silt size fraction of same sample, silt size fraction shows presence of clay minerals in addition to other mineral peaks. XRD spectrum of both clay and silt size fractions are compared in the figure 15.

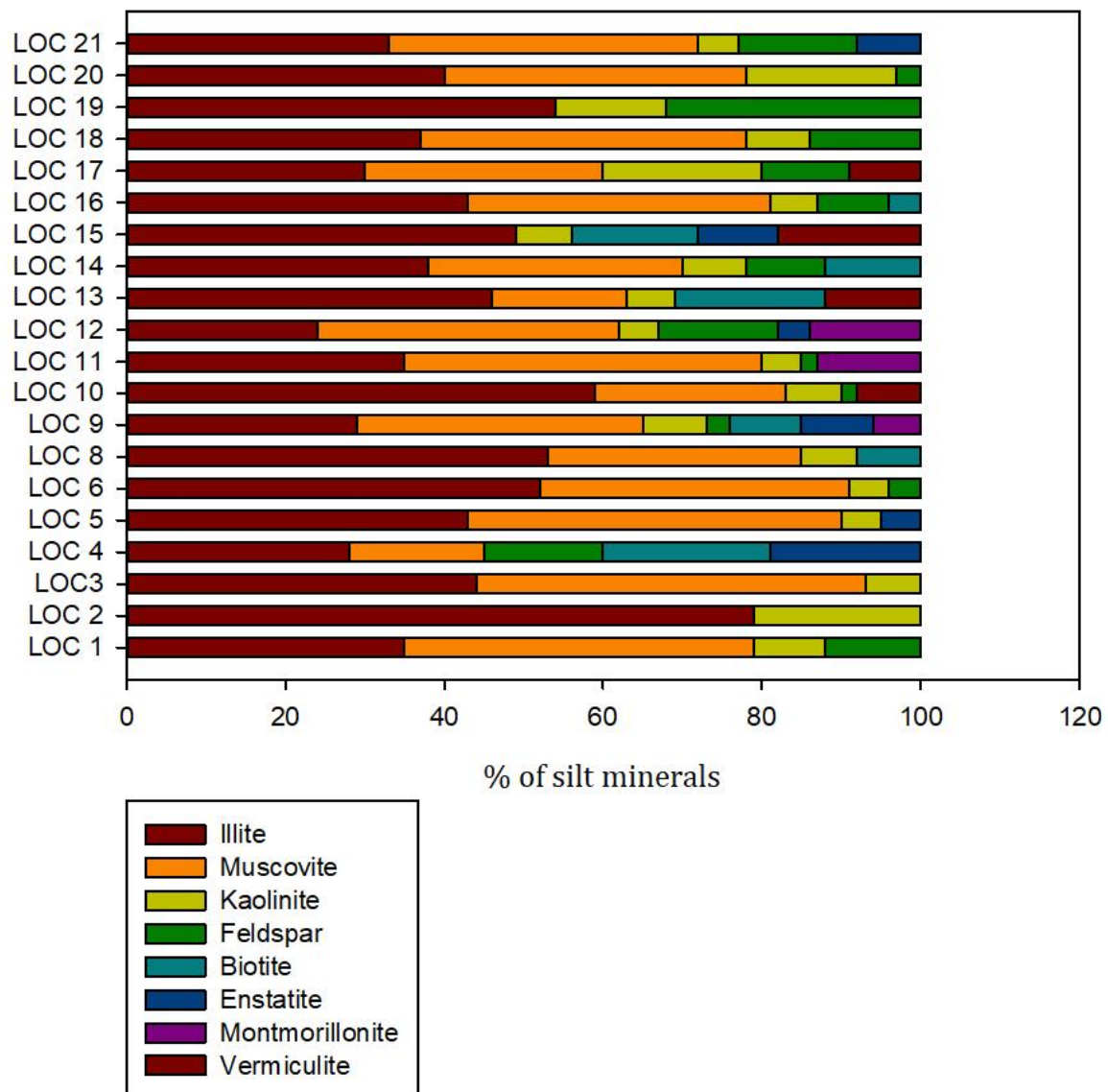


Figure 14 - Semi quantification of mineral percentage in silt

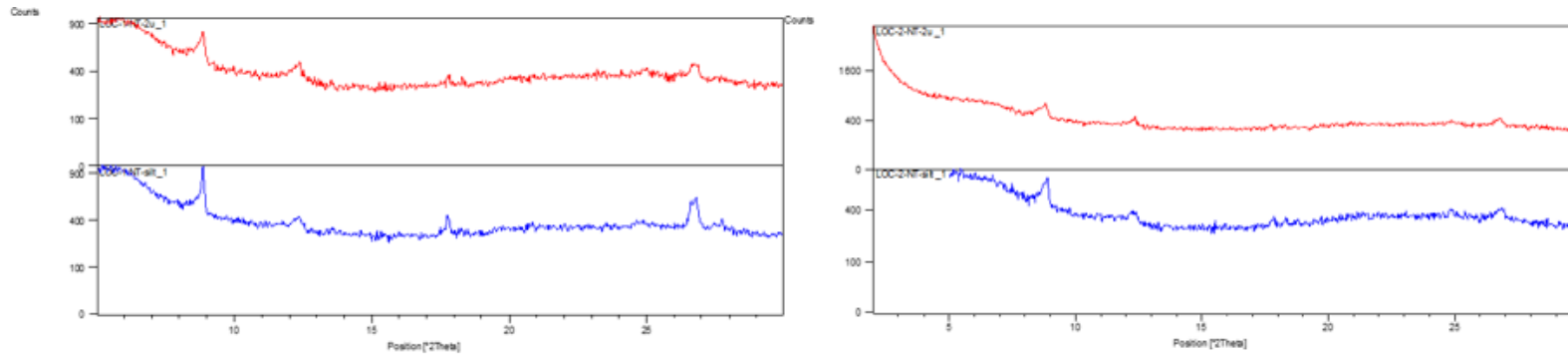


Figure 15 a) - Comparison of XRD spectrum from Location 1 & 2 (separated clay fraction is in top and silt size fraction is in the bottom).

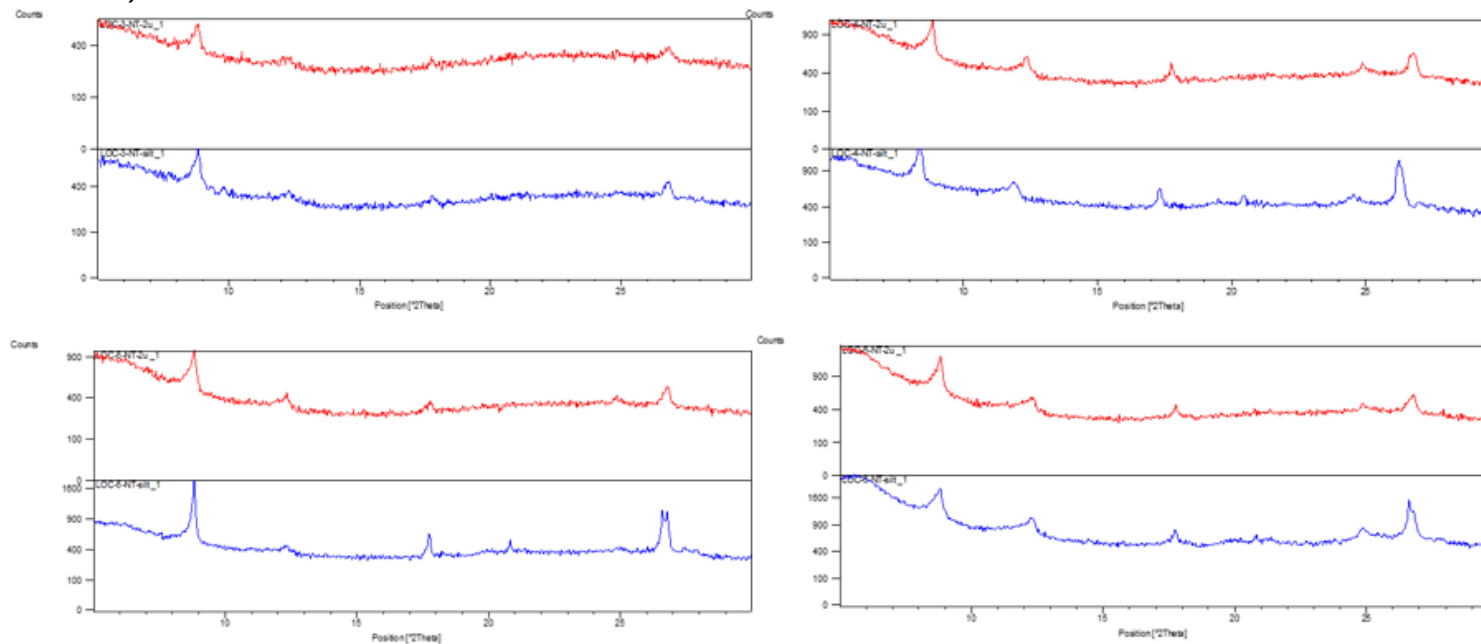


Figure 15 b) - Comparison of XRD spectrum from Location 3, 4, 5 & 6 (separated clay fraction is in top and silt size fraction is in the bottom)

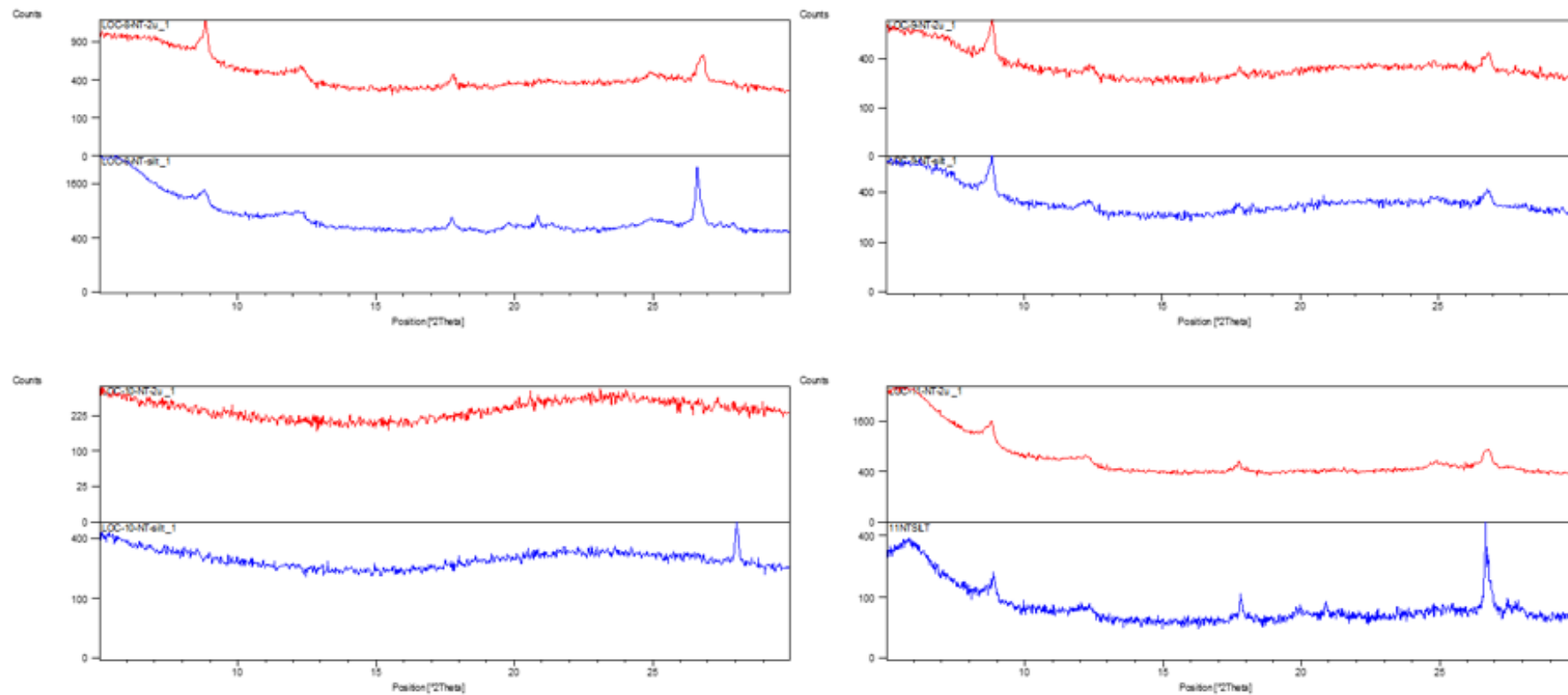


Figure 15 c) - Comparison of XRD spectrum from Location 8, 9, 10 & 11 (separated clay fraction is in top and silt size fraction is in the bottom)

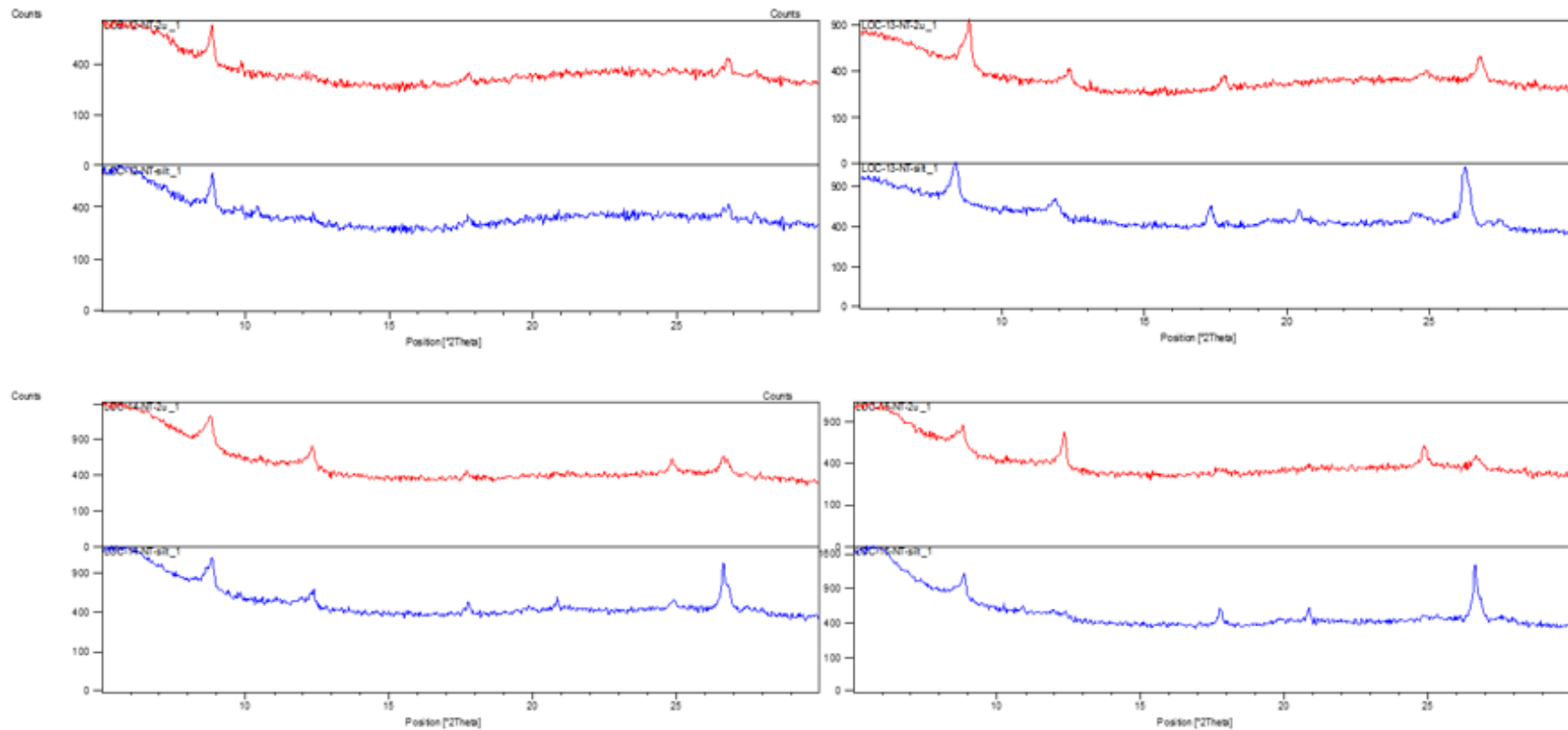


Figure 15 d) - Comparison of XRD spectrum from Location 12, 13, 14 & 15 (separated clay fraction is in top and silt size fraction is in the bottom).

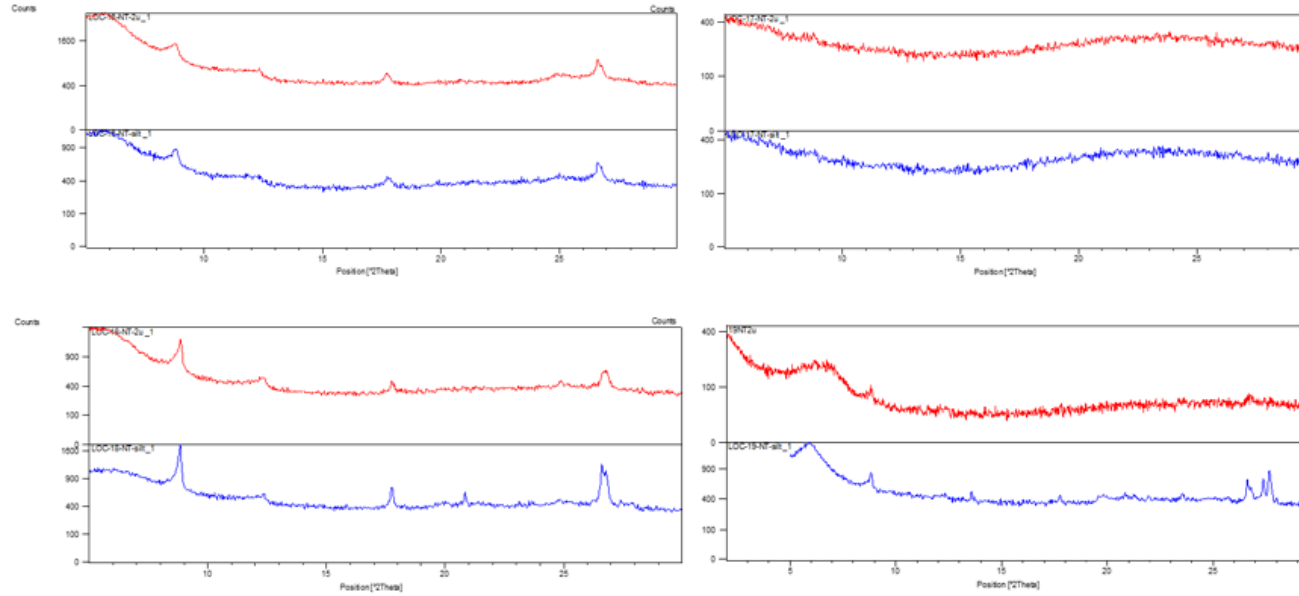


Figure 15 e) - Comparison of XRD spectrum from Location 16, 17, 18 & 19 (separated clay fraction is in top and silt size fraction is in the bottom).

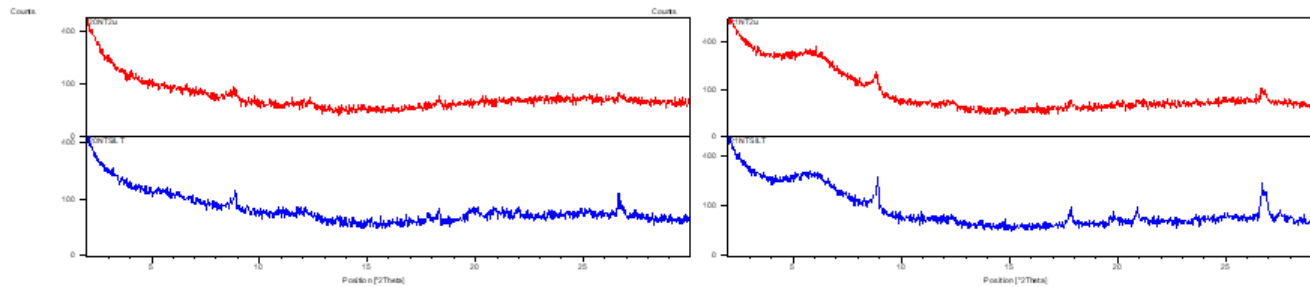
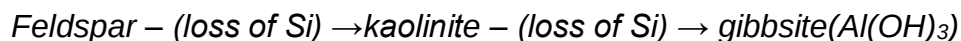


Figure 15 d) - Comparison of XRD spectrum from Location 20 & 21 (separated clay fraction is in top and silt size fraction is in the bottom).

6. Discussions

XRD analysis of oriented slide of samples shows different clay minerals, which includes kaolinite, illite, vermiculite, montmorillonite and chlorite. Other minor minerals such as muscovite and biotite are also present. In addition to the clay size fraction, silt samples were also analysed. All the minerals present in clay fraction is also found in the silt; however the semi-quantification of quartz, enstatite and feldspar are relatively high. Minor quantity of saponite, gibbsite and goethite also present in the samples. Altogether, bedload sediments shows presence of minerals likes kaolinite, illite, vermiculite, montmorillonite, chlorite, saponite, smectite, quartz, feldspar, pyroxene and olivine. Bedload sediments reflects the presence of similar secondary minerals found in the weathered profile of the catchment region. However, the semi-quantification of these minerals is different from weathering profile. The clay minerals mainly formed by loss of silica from weathering profile that are rich in alumina silicates. Kaolinite form by loss of silica and separation of Al and Fe from feldspar. The further loss of silica converts the kaolinite to gibbsites.



The bedload sediments also contain the same minerals as the above except stilbite. The clay mineral generally increases from profile to bed load sediments and their formations may be explained as follows.

During chemical weathering cations from the primary minerals are being exchanged with H^+ and release of cations like K^+ , Ca^{2+} , Na^+ etc. to aqueous phase. Further retention of cations like Al^{3+} , Si^{4+} give rise to formation of new secondary/clay minerals. This process called as new formation. For example:



Weathering of Na-feldspar (albite) to kaolinite: *albite + carbonic acid + water* → *sodium ion + bicarbonate ion + silicic acid + kaolinite*



Clay minerals in the bed load are produced by the weathering of the basalt outcrop in this region. Presence of quartz is prominent in silt samples, from loc 1 to loc 21 however the percentage is high in loc-3,4, 8,10, 11, 15 indicating intense chemical

weathering in the catchment and sediment generated from the upper layers of flow where vesicles are filled with zeolites and polymers of quartz. Saponite, carbonate, pyroxene, goethite, vermiculite, montmorillonite minerals were less in quantity, due to the high intensity of weathering. However, in some locations such as loc-9, 10, 12, 13, 14, 15, 17 and 18, vermiculite montmorillonite, are relatively higher in percentage. Samples exhibit high percentage of illite and kaolinite from loc1 to loc21 whereas pyroxene percentage is very less. However, some locations (loc-4, 5, 9, 12, 16 & 18) have high percentage due to the weathering intensity, high flow speed of water and contribution of drainage from different catchment area.

Data from bedload sediments of Gumani River (Bej, 2018) indicate that generally quartz, feldspar group mineral like albite, anorthite, orthoclase, pyroxene group minerals like enstatite, clinoenstatite, goethite synthetic, mica like muscovite, biotite, phlogopite, zeolite minerals like analcime, carbonate minerals ankerite, gibbsite, clay minerals kaolinite, illite is dominant in the area. Some magnetite, magnesite minerals were also identified. The XRD data of the bedload sediments are mainly rich in mineral like's kaolinite, pyroxene, stilbite, clinoenstatite, gibbsite and olivine. From the bedload sediments, the minerals are same and the only difference is in their semi-quantification. The clay minerals mainly formed by loss of silica from weathering profile that are rich in alumina silicates. Kaolinite formed by loss of silica and separation of Al and Fe from Feldspar. The further loss of silica converts the kaolinite to gibbsites. Clay mineral gibbsite and kaoline indicate the removal of silica is higher in the upper reach and shows kaolinitization, lateritization and up to some degree, bauxitization process is active in the upper reach. Based on CIA value, A-CN-K, A-CN-K-FM ternary diagram (Bej, 2018) indicate the intensity of weathering is moderate to high at different location.

Similar studies on clay minerals derived from basalt on Hainan Island China, shows that all of the clay fractions were dominated by kaolinite minerals. The C layers in the soils from earlier rocks had a lot of gibbsite. Illite was first formed in the A horizon of relatively young soils, then smectite mixed layer minerals (illite-smectites, then vermiculite-illite, and lastly vermiculite), and finally vermiculite. Illite's initial appearance is intriguing since these basalts lack a magmatic micaceous or phyllosilicate phase, and we attribute illite's creation to a subsequent event; perhaps brought on by alkali transfer by plant components.

From the XRD comparison data (fig-5), the mineral assemblage associated with various locations are compared with one another to understand the variability in percentage (%) of the mineral changes from one location to another. From location 1-location 21 the clay minerals are prominent. Quartz is the dominant assemblage due to its high resistance to weathering. Due to the lateralization process some iron rich minerals like Goethite also forms. Bedload sediment mainly rich in feldspar group of mineral, which undergoes weathering to form kaolinite by loss of silica and whenever there is further loss of silica the alumina-silicate mineral like gibbsite, are formed. As the provenance of the area is mainly basaltic nature, the mineral like enstatite, albite, anorthite are found abundantly. The comparison to total locations, the mineral like quartz, feldspar, clay, plagioclase are prominent with respect to their weathering index and some samples are rich in clay minerals which shows the action of weathering is higher in that location. XRD spectrum comparison figure-15 (a),(b),(c),(d) and (e), silt size fraction of the bedload sediments shows the variation in their mineralogical composition. The important minerals such as quartz, Na & Ca-plagioclase, k-feldspar, pyroxene, muscovite, phlogopite, clay minerals such as kaolinite, illite are identified from the river sediments. During the chemical weathering the least resistant minerals are weathered and resistant minerals found to be enriched in the sediments. However in the present study, minerals like Ca-rich plagioclase, pyroxene and the presence of clay minerals indicates the lower degree of chemical weathering in the catchment area.

Chapter-5

Conclusion

Based on clay mineralogical analysis of bed load sediments, following conclusions were drawn.

- The mineral compositions of the river bed load sediments from different locations shows quartz, plagioclase (albite-anorthite), pyroxene group minerals like enstatite, clinoenstatite, clay minerals such as kaolinite, illite, vermiculite, montmorillonite and clay micas, are dominant minerals. whereas orthoclase, muscovite, biotite, phlogopite are found in minor amount.
- Clay mineral percentage is higher in the upper reaches compare to the lower reach of the river.
- Clay mineral kaolinite indicate the removal of silica is higher in the upper reach and shows kaolinitization, lateritization and upto some degree of Bauxitization process is active in the upper reach.
- The index of lateritisation shows clear indication of kaolinitization → lateritization → Bauxitization process.
- As the river is draining a monolithological terrain the weathering type and its intensity shows the alteration of basaltic rock.

Future work –More XRD data of clay minerals and other minerals required to analyse and understand the process of catchment scale denudation. In addition to bed load, suspended load and dissolved load needs to be analyzed for understanding catchment scale denudation and weathering process.

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